

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

ITDS 294
Fundamentals of Electronic Commerce Systems

Session 7 - 8 : E-Commerce Marketing and Advertising



Session 7 - 8

E-Commerce Marketing and Advertising

รูปแบบการใช้งานอินเทอร์เน็ตของลูกค้า กระบวนการตัดสินใจซื้อ กลยุทธ์และเครื่องมือ Online Marketing และการโฆษณา เทคโนโลยีของ Internet Marketing ค่าใช้จ่ายและประโยชน์ของการทำ Online Marketing Communication

Learning Outcome for Session 7 - 8

E-Commerce Marketing and Advertising

นิสิตสามารถกำหนดกลยุทธ์และเลือกเครื่องมือที่ใช้ในการทำ Online Marketing และการทำโฆษณาให้กับ E-Commerce /E-Business ที่ต้องการพัฒนา



E-Commerce Marketing and Advertising

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Consumer Online : The Internet Audience and Consumer Behavior

Before firms can begin to sell their products online, they must first understand what kinds of people they will find online and how those people behave in the online marketplace. In this section, we focus primarily on individual consumers in the business- to-consumer (B2C) arena. However, many of the factors discussed apply to the B2B arena as well, insofar as purchasing decisions by firms are made by individuals.



Internet Traffic Patterns The Online Consumer Profile

TABLE 6.1 WHAT'S NEW IN ONLINE MARKETING AND ADVERTISING 2020–2021

BUSINESS
• Although online marketing and advertising spending had been increasing by over 20% a year over the last five years (while most forms of traditional media marketing and advertising were either flat or declining), the Covid-19 pandemic impacted all types of ad spending, including digital, in 2020. Digital ad spending worldwide increased by less than 5% in 2020 but is expected to resume growing at double-digit rates in 2021 and continue to comprise an ever-increasing share of all ad spending, reaching almost 65% by 2024. • Mobile advertising is the platform for over 70% of all digital advertising spending. • Search engine marketing and advertising remains important, but its rate of growth is slowing somewhat compared to other formats. Search ad spending remained relatively flat in 2020 due to the Covid-19 pandemic, but is expected to resume growing at a faster pace in 2021. • Digital video ads remain the fastest-growing ad format. • Social network advertising and marketing spending continues to expand. • Viewability issues and ad fraud continue to cause concerns for marketers. • Native advertising and other forms of content marketing rise.
TECHNOLOGY
• Ad blocking software creates concern for both online publishers and advertisers. • Big data: online tracking produces oceans of data, challenging business analytics programs. • Cloud computing makes rich marketing content and multi-channel, cross-platform marketing a reality. • Programmatic advertising (an automated, technology-driven method of buying and selling display and video ads) takes off.
SOCIETY
• Targeted advertising based on behavioral tracking leads to growing privacy awareness and fears. • Marketers become increasingly concerned about placement of their ads next to controversial online content, giving rise to advertising boycotts and companies focusing on brand safety.

Internet Traffic Patterns The Online Consumer Profile

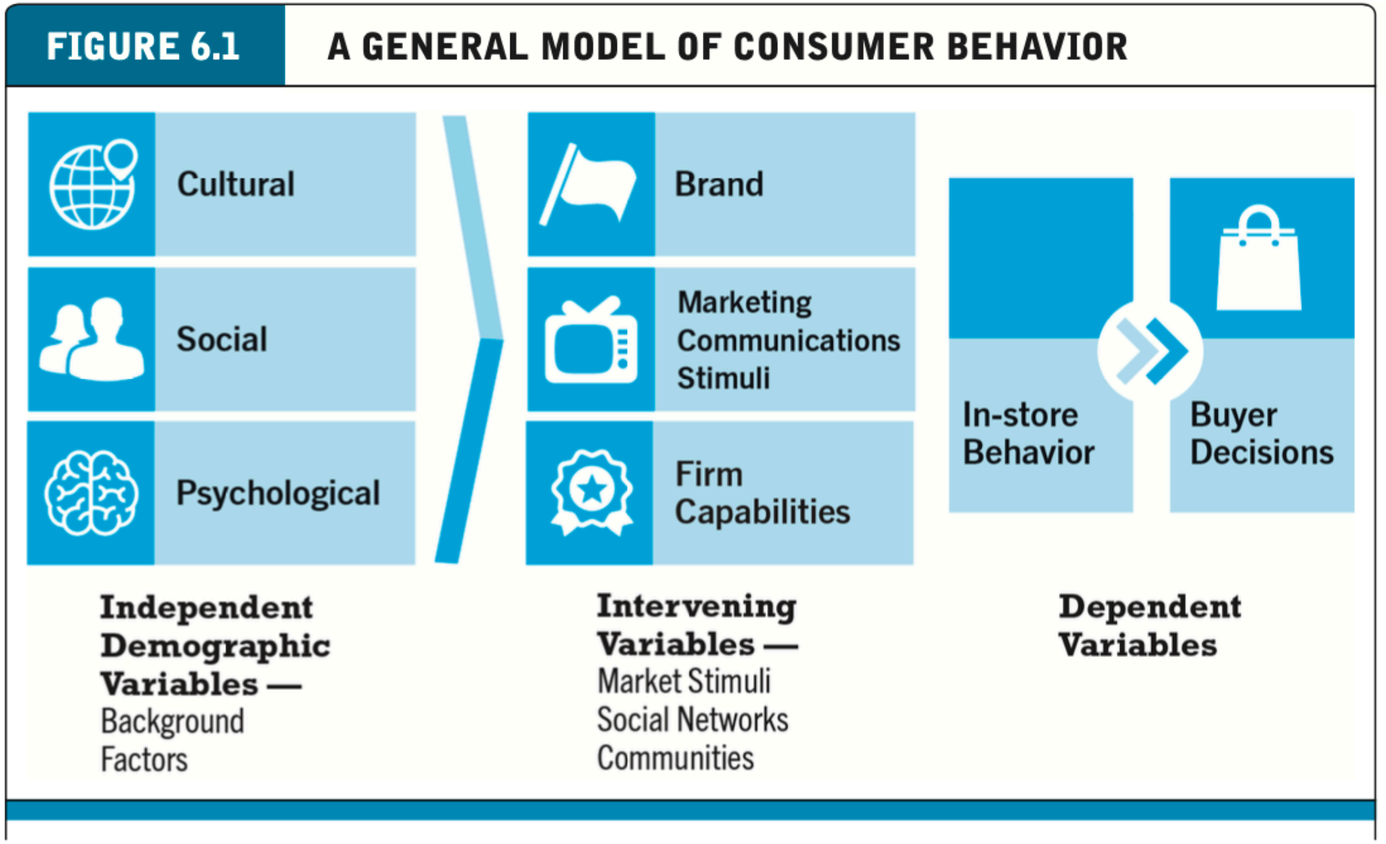
Let see digital adoption and use in Thailand of 2026



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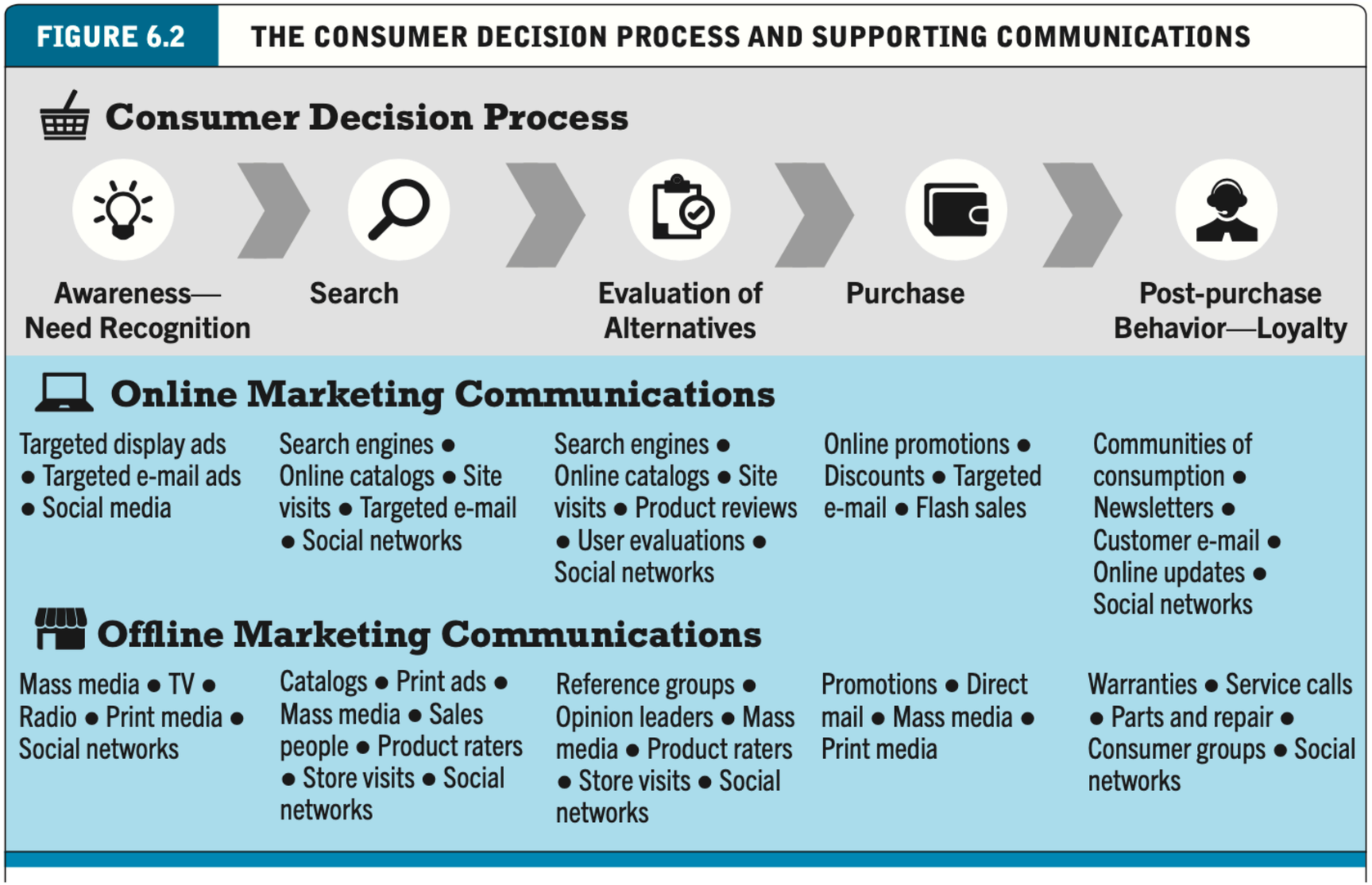
Customer Behavior Models

Once firms have an understanding of who is online, they need to focus on how consumers behave online. The study of consumer behavior is a social science discipline that attempts to model and understand the behavior of humans in a marketplace. Several social science disciplines play roles in this study, including sociology, psychology, and economics.



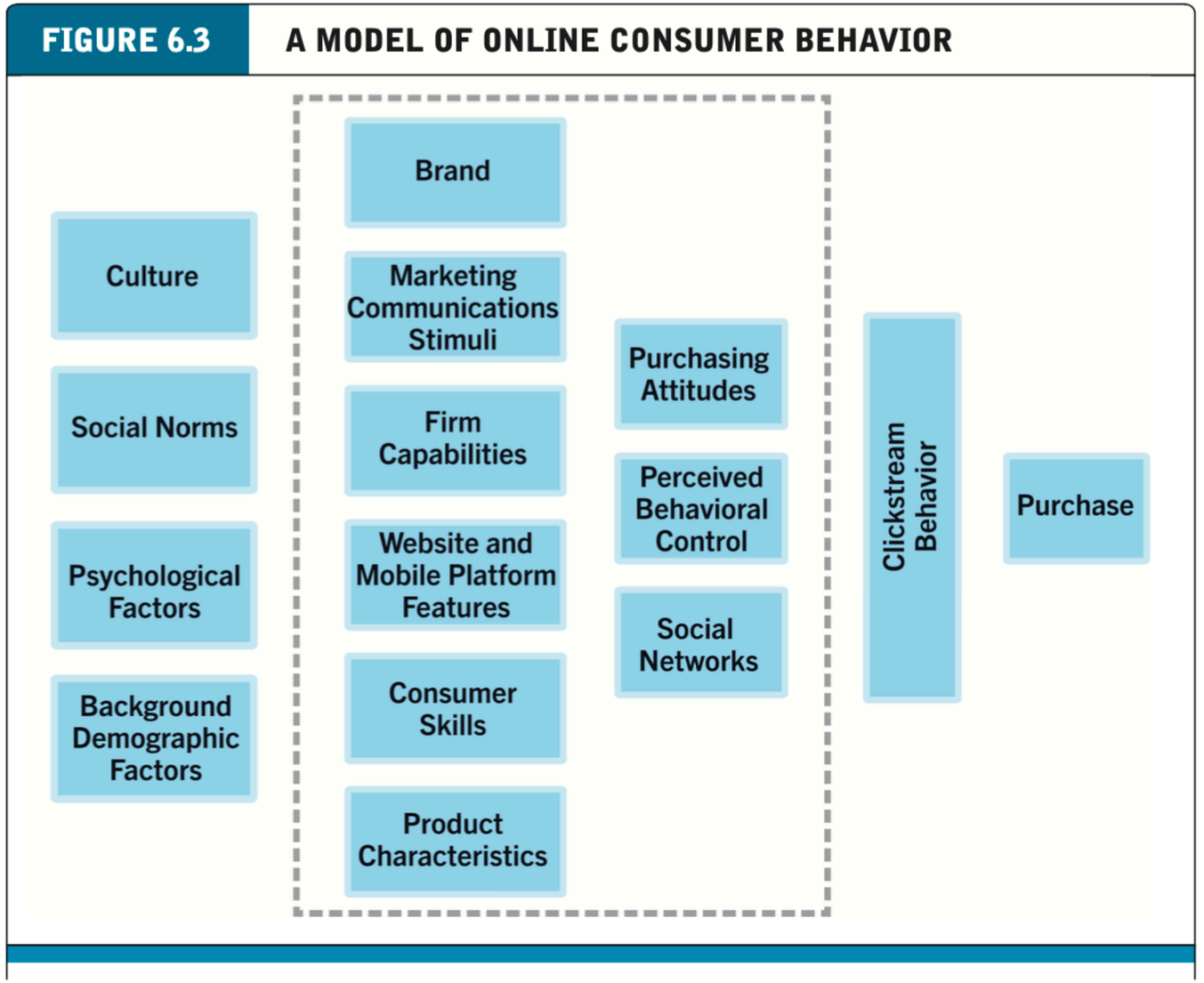
The Online Purchasing Decision

FIGURE 6.2 THE CONSUMER DECISION PROCESS AND SUPPORTING COMMUNICATIONS



The Online Purchasing Decision

In the online model, website and mobile platform features, along with consumer skills, product characteristics, attitudes toward online purchasing, and perceptions about control over the online environment come to the fore. Website and mobile platform features include latency (delay in downloads), navigability, and confidence in online security.



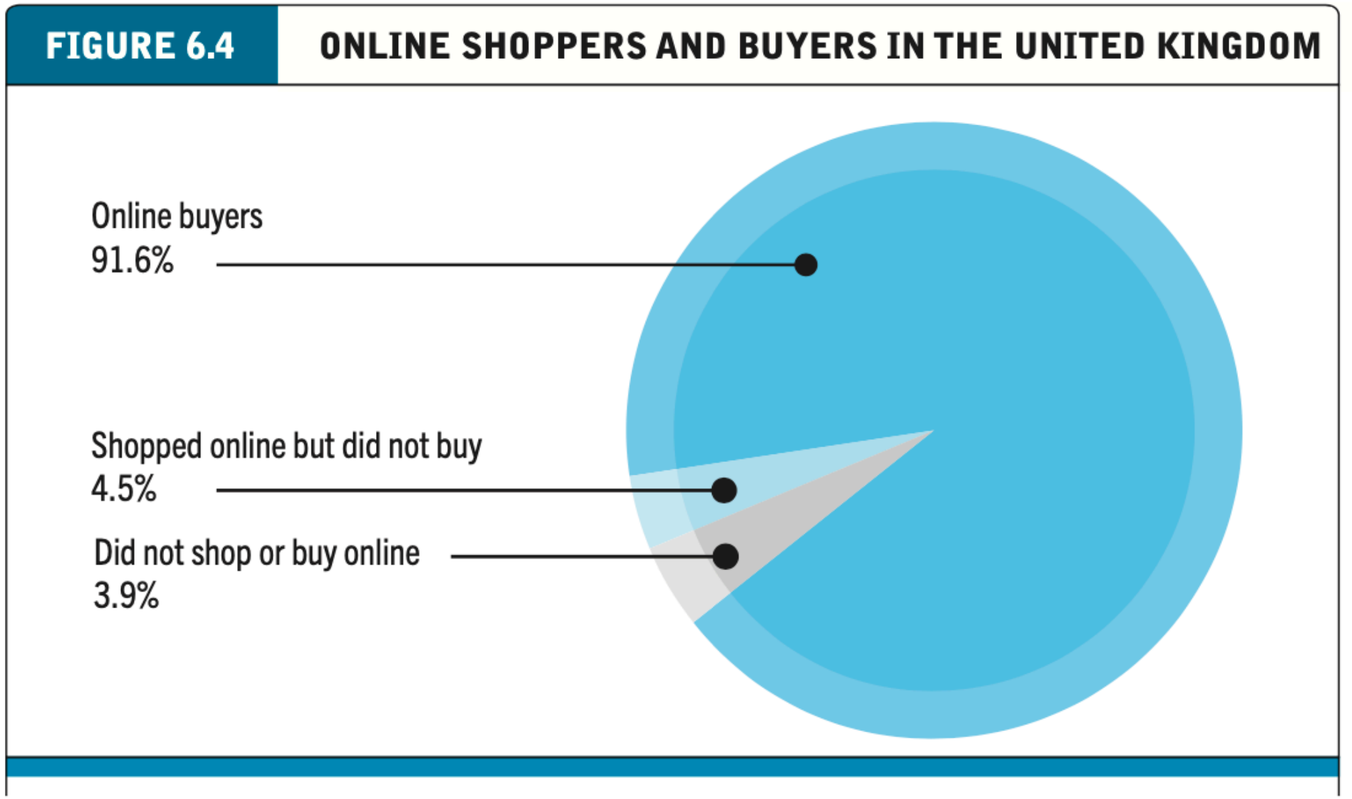
The Online Purchasing Decision

Clickstream behavior refers to the transaction log that consumers establish as they move about the Web, from search engine to a variety of sites, then to a single site, then to a single page, and then, finally, to a decision to purchase.

A study of over 10,000 visits to an online wine store found that detailed and general clickstream behavior were as important as customer demographics and prior purchase behavior in predicting a current purchase.

Shoppers : Browsers and Buyers

E-commerce is a major conduit and generator of offline commerce. The reverse is also true: online traffic is driven by offline brands and shopping. While online research influences offline purchases, it is also the case that offline marketing media heavily influence online behavior including sales.



Traditional print media (magazines and newspapers) and television are by far the most powerful media for reaching and engaging consumers with information about new products and directing them to the Web.

Shoppers : Browsers and Buyers

You can look at online sales as divided roughly into two groups: small-ticket and big-ticket items. Big-ticket items include computer equipment and consumer electronics, where orders can easily be more than $1,000. Small-ticket items include apparel, books, health and beauty supplies, office supplies, music, software, videos, and toys, where the average purchase is typically less than $100.

In the early days of e-commerce, sales of small-ticket items vastly outnumbered those of large-ticket items. But the recent growth of big-ticket items such as computer hardware, consumer electronics, furniture, and jewelry has changed the overall sales mix. Consumers are now much more confident spending online for big-ticket items.

Although furniture and large appliances were ini- tially perceived as too bulky to sell online, these categories have rapidly expanded in the last few years.

Online Marketing and Advertising Strategies and Tools

Online marketing has many similarities to, and differences from, ordinary marketing. The objective of online marketing—as in all marketing—is to build customer relation- ships so that the firm can achieve above-average returns (both by offering superior products or services and by communicating the brand's features to the consumer).



Online Marketing and Advertising Strategies and Tools

The five main elements of a comprehensive multi-channel marketing plan are: website, traditional online marketing, social marketing, mobile marketing, and offline marketing. **Table 6.2** illustrates these five main platforms, central elements within each type, some examples, and the primary function of marketing in each situation.

TABLE 6.2 THE DIGITAL MARKETING ROADMAP

TYPE OF MARKETING	PLATFORMS	EXAMPLES	FUNCTION
Website	Traditional website	Ford.com	Anchor site
Traditional Online Marketing	Search engine marketing	Google; Bing; Yahoo	Query-based intention marketing
	Display advertising	Yahoo; Google; MSN	Interest- and context-based marketing; targeted marketing
	E-mail	Major retailers	Permission marketing
	Affiliates	Amazon	Brand extension
Social Marketing	Social networks	Facebook	Conversations; sharing
	Micro blogging sites	Twitter	News, quick updates
	Blogs/forums	Tumblr	Communities of interest; sharing
	Visual marketing	Pinterest/Instagram	Branding; sharing
	Video marketing	YouTube	Engage; inform
	Game marketing	Chipotle Spot the Imposter game	Identification
Mobile Marketing	Mobile site	m.ford.com	Quick access; news; updates
	Apps	Ford Mustang Customizer app	Visual engagement
		My Ford	Visual engagement
Offline Marketing	Television	Apple/The Human Family: Shot on iPhone	Brand anchoring; inform
	Newspapers	American Airlines/ The World's Greatest Flyers Fly American	Brand anchoring; inform
	Magazines	Apple Watch/Vogue Magazine	Brand anchoring; inform

The Website as a Marketing Platform:
Establishing the Customer Relationship

The website performs four important functions: establishing the brand identity and consumer expectations, informing and educating the consumer, shaping the customer experience, and anchoring the brand in an ocean of marketing messages coming from different sources.

Customer experience refers to the totality of experiences that a customer has with a firm, including the search, informing, purchase, consumption, and after-sales support for the product.

Traditional Online Marketing and Advertising Tools

Online Advertising

- An online advertising refers to a paid message on a website, app, or other digital medium
- Online advertising includes display ads (banners, video, rich media, and sponsorships), search ads, classifieds, lead generation, and e-mail, on desktop and laptop computers, as well as mobile devices and connected TVs.

TABLE 6.3 ONLINE ADVERTISING SPENDING IN THE UNITED STATES FOR SELECTED FORMATS (IN BILLIONS)

FORMAT	2020	2024	AVERAGE GROWTH RATE
Search	$54.4	$83.6	8.6%
Banner ads	$31.1	$54.1	12.1%
Video	$35.5	$69.4	17%
Rich media	$5.6	$8.0	8%
Sponsorships	$2.8	$3.8	6.3%
Lead generation	$2.5	$3.3	5.2%
Classifieds	$2.1	$2.5	5.5%
E-mail	$0.49	$0.67	6.7%

Traditional Online Marketing and Advertising Tools

Search Engine Marketing and Advertising

- **Search engine marketing** (**SEM**) refers to the use of search engines to build and sustain brands.
- **Search engine advertising** refers to the use of search engines to sup- port direct sales to online consumers.
  - **Organic search** inclusion and ranking of sites depends on a more or less unbiased application of a set of rules imposed by the search engine
  - **Paid inclusion** for a fee, guarantees a website's inclusion in its list of sites, more frequent visits by its web crawler, and suggestions for improving the results of organic searching
    - In **keyword advertising,** merchants purchase keywords through a bidding process at search sites, and whenever a consumer searches for that word, their advertisement shows up somewhere on the page, usually as a small text-based advertisement on the right, but also as a listing on the very top of the page.
    - **Network keyword advertising** (**context advertising**), introduced by Google as its AdSense product in 2002, differs from the ordinary keyword advertising described previously. Publishers (websites that want to show ads) join these networks and allow the search engine to place "relevant" ads on their sites.

Traditional Online Marketing and Advertising Tools

Search Engine Marketing and Advertising

- While search engines have provided significant benefits to mer- chants and customers, they also present risks and costs
  - **Click fraud** occurs when a competitor clicks on search engine results and ads, forcing the advertiser to pay for the click even though the click is not legitimate.
  - **Content farms** are companies that generate large volumes of textual content for multiple websites designed to attract viewers and search engines.
  - **Link farms** are groups of websites that link to one another, thereby boosting their ranking in search engines that use a page ranking algorithm to judge the “usefulness” of a site.

Traditional Online Marketing and Advertising Tools

Display Ad Marketing

- Display ads include a number of different types of ads, including banner ads, rich media ads, and video ads.
  - **Banner ads** are the oldest and most familiar form of display marketing. They are also the least-effective and the lowest-cost form of online marketing.
  - **Rich Media Ads** refers to Ads that employ interactive features that engage the user, such as ani- mations (moving graphics), or elements that trigger new content experiences, such as ad expansion, where the ad expands to a size bigger than its original size, or video play
  - **Video Ads** are TV-like advertisements that appear as in-page video commercials before, during, or after a variety of content.
  - **Sponsorship** is a paid effort to tie an advertiser's name to particular information, an event, or a venue in a way that reinforces its brand in a positive yet not overtly commercial manner. Sponsorships typically are more about branding than immediate sales.
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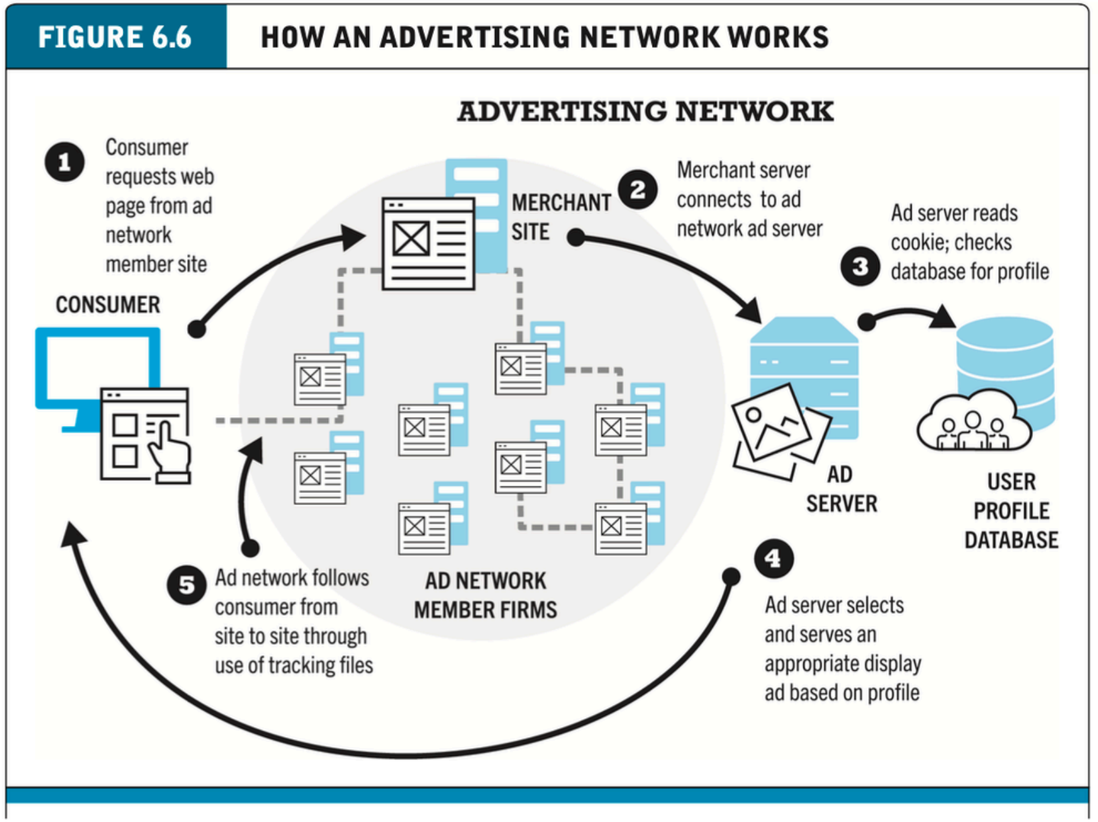
Traditional Online Marketing and Advertising Tools

Content Marketing and Advertising Network

- **Content marketing** creates a content campaign for a brand and then tries to secure placement on a variety of websites. Examples of content include articles, info- graphics, case studies, interactive graphics, white papers, and even traditional press releases. The aim of content marketing is to increase visitors to a company's website, organic search rankings, and brand engagement via social media.
- **Advertising networks** appeared to help firms take advantage of the powerful marketing potential of the Internet, and to make the entire process of buying and selling online ads more efficient and transparent.

Traditional Online Marketing and Advertising Tools

Content Marketing and Advertising Network



Traditional Online Marketing and Advertising Tools

Display Advertising Issues

- **Ad fraud** : The online advertising industry has become increasingly concerned about the issue of ad fraud. Ad fraud involves the practice of falsifying web or mobile traffic in order to charge advertisers for impressions, clicks, or other actions that never actually occurred.
- **Viewability:** A significant percentage of online ad impressions served are not actually viewed or even viewable

Traditional Online Marketing and Advertising Tools

E-mail Marketing

- When e-mail marketing began, unsolicited e-mail was not common.
- Direct e-mail marketing (e-mail marketing messages sent directly to interested users) was one of the first and most effective forms of online marketing communications. Direct e-mail marketing messages are sent to an opt-in audience of Internet users who, at one time or another, have expressed an interest in receiving messages from the advertiser.
- By sending e-mail to an opt-in audience, advertisers are targeting interested consumers. By far, in-house e-mail lists are more effective than purchased e-mail lists. Because of the comparatively high response rates and low cost, direct e-mail marketing remains a common form of online marketing communications.
- Other benefits of e-mail marketing include its mass reach, the ability to track and measure response, the ability to personalize content and tailor offers, the ability to drive traffic to websites for more interaction, the ability to test and optimize content and offers, and the ability to target by region, demographic, time of day, or other criteria.

Traditional Online Marketing and Advertising Tools

Affiliate Marketing

- Affiliate marketing is a form of marketing where a firm pays a commission, typically anywhere between 4% to 20%, to other websites (including blogs) for sending customers to their website. Affiliate marketing generally involves pay-for-performance: the affiliate or affiliate network gets paid only if users click on a link or purchase a product.
- Visitors to an affiliate website typically click on ads and are taken to the advertiser's website. In return, the advertiser pays the affiliate a fee, either on a per-click basis or as a percentage of whatever the customer spends on the advertiser's site. Paying commissions for referrals or recommendations long predated the Web.
- For instance, Amazon has the world's largest affiliate program, called Amazon Associates. Participant sites receive up to 10% commission on sales their referrals generate. Affiliates attract people to their blogs or websites where they can click on ads for products at Amazon.

Traditional Online Marketing and Advertising Tools

Viral Marketing

- Just as affiliate marketing involves using a trusted website to encourage users to visit other sites, **viral marketing** is a form of social marketing that involves getting customers to pass along a company's marketing message to friends, family, and colleagues.
- It's the online version of word-of-mouth advertising, which spreads even faster and further than in the real world. In the offline world, next to television, word of mouth is the second most important means by which consumers find out about new products.
- And the most important factor in the decision to purchase is the **face-to-face recommendations** of parents, friends, and colleagues.
- Millions of online adults in the United States are "influencers" who share their opinions about products in a variety of online settings. In addition to increasing the size of a company's customer base, customer referrals also have other advantages: they are less expensive to acquire because existing customers do all the acquisition work, and they tend to use online support services less, preferring to turn back to the person who referred them for advice.

Social, Mobile, and Local Marketing and Advertising

Social marketing and advertising involves the use of online social networks and communities to build brands and drive sales revenues. There are several kinds of social networks, from Facebook, Twitter, Pinterest, and Instagram, to social apps, social games, blogs, and forums (websites that attract people who share a community of interests or skills).

Social networks offer advertisers all the main advertising formats, including ban- ner ads, native advertising, short pre-roll and post-roll ads associated with videos, and sponsorship of content. Having a corporate Facebook page is in itself a marketing tool for brands just like a web page. Many firms, such as Coca-Cola, have shut down product-specific web pages and instead use Facebook pages.

Social, Mobile, and Local Marketing and Advertising

Blogs can also be used for social marketing. Blogs have been around for a decade and are a part of the mainstream online culture. Because blog readers and creators tend to be more educated, have higher incomes, and be opinion leaders, blogs are ideal platforms for ads for many products and services that cater to this kind of audience. Because blogs are based on the personal opinions of the writers, they are also an ideal platform to start a viral marketing campaign.

Influencer marketing is another form of social media marketing that began initially with bloggers and has since expanded to a wide variety of social networks. Influencer marketing uses endorsements and product mentions from people who have dedicated followers on social media and who are viewed by those followers as trusted experts or celebrities. Brands seek to leverage the trust influencers have built up with their following and translate recommendations from influencers into sales.

Social, Mobile, and Local Marketing and Advertising

Mobile marketing includes the use of display banner ads, rich media, video, native advertising, games, e-mail, text messaging, in-store messaging, Quick Response (QR) codes, and couponing. Mobile is now a required part of the standard marketing budget.

Along with social marketing and mobile marketing, **local marketing** is the third major trend in e-commerce marketing in 2020–2021. The growth of mobile devices has accelerated the growth of local search and purchasing. New marketing tools like local advertisements on social networks and daily deal sites are also contributing to local marketing growth.

Other Online Marketing Strategies

Personalization, One-to-One Marketing, and Interest-based Advertising (Behavioral Targeting)

No Internet-based marketing technique has received more popular and academic comment than “one-to-one” or “personalized marketing.” One-to-one mar- keting (personalization) segments the market on the basis of individuals (not groups), based on a precise and timely understanding of their needs, targeting specific marketing messages to these individuals.

One-to-one marketing is the ultimate form of market segmentation, targeting, and positioning—where the segments are individuals.

Other Online Marketing Strategies

Personalization, One-to-One Marketing, and Interest-based Advertising (Behavioral Targeting)

Behavioral targeting involves using the online and offline behavior of consumers to adjust the advertising messages delivered to them online, often in real time (milliseconds from the consumer's first URL entry). The intent is to increase the efficiency of marketing and advertising, and to increase the revenue streams of firms who are in a position to behaviorally target visitors. Because behavioral targeting as a label has somewhat unfavorable connotations, the online advertising industry, led by Google, has introduced a different name for behavioral targeting. They call it **interest-based advertising** (**IBA**).

Other Online Marketing Strategies

Customer Services

Frequently asked questions (**FAQs**), a text-based listing of common questions and answers, provide an inexpensive way to anticipate and address customer concerns. Adding a FAQ page on a website linked to a search engine helps users track down needed information more quickly, enabling them to help themselves resolve questions and concerns. By directing customers to the FAQs page first, websites can give customers answers to common questions. If a question and answer do not appear, it is important for sites to make contacting a live person simple and easy. Offering an e-mail link to customer service at the bottom of the FAQs page is one solution.

Other Online Marketing Strategies

Customer Services

Real-time customer service chat systems (in which a company's customer service representatives interactively exchange text-based messages with one or more customers on a real-time basis) are an increasingly popular way for companies to assist online shoppers during a purchase. Chats with online customer service representatives can provide direction, answer questions, and troubleshoot technical glitches that can kill a sale.

Intelligent agent technology is another way to provide assistance to online shop- pers. Intelligent agents are part of an effort to reduce costly contact with customer service representatives. Automated response systems send e-mail order confirmations and acknowledgments of e-mailed inquiries, in some cases letting the customer know that it may take a day or two to actually research an answer to their question. Automated shipping confirmations and order status reports are also common.

Internet Marketing Technologies

Internet marketing has many similarities to and differences from ordinary marketing. The objective of Internet marketing—as in all marketing—is to build customer relationships so that the firm can achieve above-average returns (both by offering superior products or services and by communicating the product's features to the consumer).



Internet Marketing Technologies

This table describes how marketing has changed as a result of these new technical capabilities.

TABLE 6.5 IMPACT OF UNIQUE FEATURES OF E-COMMERCE TECHNOLOGY ON MARKETING

E-COMMERCE TECHNOLOGY DIMENSION	SIGNIFICANCE FOR MARKETING
Ubiquity	Marketing communications have been extended to the home, work, and mobile platforms; geographic limits on marketing have been reduced. The marketplace has been replaced by "marketspace" and is removed from a temporal and geographic location. Customer convenience has been enhanced, and shopping costs have been reduced.
Global reach	Worldwide customer service and marketing communications have been enabled. Potentially hundreds of millions of consumers can be reached with marketing messages.
Universal standards	The cost of delivering marketing messages and receiving feedback from users is reduced because of shared, global standards of the Internet.
Richness	Video, audio, and text marketing messages can be integrated into a single marketing message and consuming experience.
Interactivity	Consumers can be engaged in a dialog, dynamically adjusting the experience to the consumer, and making the consumer a co-producer of the goods and services being sold.
Information density	Fine-grained, highly detailed information on consumers' real-time behavior can be gathered and analyzed for the first time. "Data mining" Internet technology permits the analysis of terabytes of consumer data every day for marketing purposes.
Personalization/ Customization	This feature potentially enables product and service differentiation down to the level of the individual, thus strengthening the ability of marketers to create brands.
Social technology	User-generated content and social networks, along with blogs, have created large new audiences online, where the content is provided by users. These audiences have greatly expanded the opportunity for marketers to reach new potential customers in a nontraditional media format. Entirely new kinds of marketing techniques are evolving. These same technologies expose marketers to the risk of falling afoul of popular opinion by providing more market power to users who now can "talk back."

Internet Marketing Technologies

WEB TRANSACTION LOGS

A primary source of information about online consumers is the transaction log maintained by all web servers. A transaction log records user activity at a website. The transaction log is built into web server software. Transaction log data becomes even more useful when combined with two other visitor- generated data trails: registration forms and the shopping cart database. For instance, here are just a few of the interesting marketing questions that can be answered by examining a site's web transaction logs, registration forms, and shopping cart database:

Internet Marketing Technologies

WEB TRANSACTION LOGS

- What are the major patterns of interest and purchase for groups and individuals?
- After the home page, where do most users go first, and then second and third?
- What are the interests of specific individuals (those we can identify)?
- How can we make it easier for people to use our site so they can find what they want?
- How can we change the design of the site to encourage visitors to purchase our high-margin products?
- Where are visitors coming from (and how can we optimize our presence on these referral sites)?
- How can we personalize our messages, offerings, and products to individual users?

Internet Marketing Technologies

SUPPLEMENTING THE LOGS: COOKIES AND OTHER TRACKING FILES

While transaction logs create the foundation of online data collection at a single website, marketers use tracking files to follow users across the Web as they visit other sites.

A **cookie** is a small text file that websites place on the hard disk of visitors' client computers every time they visit, and during the visit, as specific pages are visited. Cookies allow a website to store data on a user's computer and then later retrieve it. The cookie typically includes a name, a unique ID number for each visitor that is stored on the user's computer, the domain (which specifies the web server/domain that can access the cookie), a path (if a cookie comes from a particular part of a website instead of the main page, a path will be given), a security setting that provides whether the cookie can only be transmitted by a secure server, and an expiration date (not required).

Internet Marketing Technologies

SUPPLEMENTING THE LOGS: COOKIES AND OTHER TRACKING FILES

A cookie provides web marketers with a very quick means of identifying the customer and understanding his or her prior behavior at the site.

Websites use cookies to determine how many people are visiting the site, whether they are new or repeat visitors, and how often they have visited, although this data may be somewhat inaccurate because people share computers, they often use more than one computer, and cookies may have been inadvertently or intentionally erased.

Cookies make shopping carts and “quick checkout” options possible by allowing a site to keep track of a user as he or she adds to the shopping cart. Each item added to the shopping cart is stored in the site’s database along with the visitor’s unique ID value.

Internet Marketing Technologies

SUPPLEMENTING THE LOGS: COOKIES AND OTHER TRACKING FILES

Although cookies are site-specific (a website can only receive the data it has stored on a client computer and cannot look at any other cookie), when combined with **web beacons** (also called “bugs,” “clear GIFs,” or “pixel tags”), they can be used to create cross- site profiles.

Web beacons are tiny (1-pixel) graphic files embedded in e-mail messages and on websites. Web beacons are used to automatically transmit information about the user and the page being viewed to a monitoring server in order to collect personal browsing behavior and other personal information.

Internet Marketing Technologies

SUPPLEMENTING THE LOGS: COOKIES AND OTHER TRACKING FILES

In an effort to more effectively track consumers across devices, other cross-device tracking methods have begun to be developed.

Deterministic cross-device tracking relies on personally identifiable information such as an e-mail address used to log into an app and website on different devices. Facebook, Google, Apple, Twitter, and other companies that have very large user bases and have both desktop and mobile proper- ties that require logins are the most likely to be able to effectively exploit deterministic matching.

Probabilistic cross-device tracking uses algorithms to analyze thousands of anonymous data points, such as device type, operating system, and IP address, to create a possible match. This type of matching is, not surprisingly, less accurate than deterministic matching

Internet Marketing Technologies

DATABASES, DATA WAREHOUSES, DATA MINING, AND BIG DATA

Databases, data warehouses, data mining, and the variety of marketing decision-making techniques loosely called *profiling* are at the heart of the revolution in Internet market- ing.

Profiling uses a variety of tools to create a digital image for each consumer. This image can be quite inexact, even primitive, but it can also be as detailed as a character in a novel. The quality of a consumer profile depends on the amount of data used to create it, and the analytical power of the firm's software and hardware. Together, these techniques attempt to identify precisely who the online customer is and what they want, and then, to fulfill the customer's criteria exactly.

Internet Marketing Technologies

DATABASES, DATA WAREHOUSES, DATA MINING, AND BIG DATA

A **database** is a software application that stores records and attributes. A telephone book is a physical database that stores records of individuals and their attributes such as names, addresses, and phone numbers. A **database management system** (**DBMS**) is a software application used by organizations to create, maintain, and access databases. The most common DBMS are DB2 from IBM and a variety of SQL data- bases from Oracle, Sybase, and other providers. **Structured query language** (**SQL**) is an industry-standard database query and manipulation language used in relational data- bases. **Relational databases** such as DB2 and SQL represent data as two-dimensional tables with records organized in rows, and attributes in columns, much like a spread- sheet. The tables—and all the data in them—can be flexibly related to one another as long as the tables share a common data element.

Internet Marketing Technologies

DATABASES, DATA WAREHOUSES, DATA MINING, AND BIG DATA

A data warehouse is a database that collects a firm's transactional and customer data in a single location for offline analysis by marketers and site managers. The data originate in many core operational areas of the firm, such as website transaction logs, shopping carts, point-of-sale terminals (product scanners) in stores, warehouse inventory levels, field sales reports, external scanner data supplied by third parties, and financial payment data.

The purpose of a data warehouse is to gather all the firm's transaction and customer data into one logical repository where it can be analyzed and modeled by managers without disrupting or taxing the firm's primary transactional systems and databases.

Internet Marketing Technologies

DATABASES, DATA WAREHOUSES, DATA MINING, AND BIG DATA

Data mining is a set of analytical techniques that look for patterns in the data of a database or data warehouse, or seek to model the behavior of customers. Website data can be "mined" to develop profiles of visitors and customers. A **customer profile** is simply a set of rules that describe the typical behavior of a customer or a group of customers at a website. Customer profiles help to identify the patterns in group and individual behavior that occur online as millions of visitors use a firm's website.

Internet Marketing Technologies

DATABASES, DATA WAREHOUSES, DATA MINING, AND BIG DATA

The term **big data** refers to this avalanche of digital data that creates huge data sets, often from different sources, in the petabyte and exabyte range. The volumes of data are so large that traditional DBMS cannot capture, store, and analyze the data in a reasonable time.

Marketers are interested in big data because it allows them to link huge amounts of data from a variety of different sources, which in the past they were unable to do, and mine it for patterns of consumer behavior, with the potential to provide new insights into customer behavior, financial market activity, or other phenomena.

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DATABASES, DATA WAREHOUSES, DATA MINING, AND BIG DATA

To handle unstructured and semi-structured data in vast quantities, as well as structured data, many organizations are using Hadoop.

Hadoop is an open source software framework managed by the Apache Software Foundation that enables distributed parallel processing of huge amounts of data across inexpensive computers. It breaks a big data problem down into subproblems, distributes them among up to thousands of inexpensive computer processing nodes, and then combines the result into a smaller data set that is easier to analyze.

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MARKETING AUTOMATION AND CUSTOMER RELATIONSHIP MANAGEMENT (CRM) SYSTEMS

Marketing automation systems are software tools that marketers use to track all the steps in the lead generation part of the marketing process. The marketing process begins with making the potential customer aware of the firm and product and recognizing the need for the product. From there, consumers need to find you as they search for products; they will compare your products with your competitors' offerings and at some point, choose to purchase.

Software can help in each of these stages of the marketing process. A number of firms sell software packages that can visualize most of the online marketing activities of a firm and then track the progression from exposure to display ads, finding your firm on a search engine, directing follow-up e-mail and communications, and finally a purchase.

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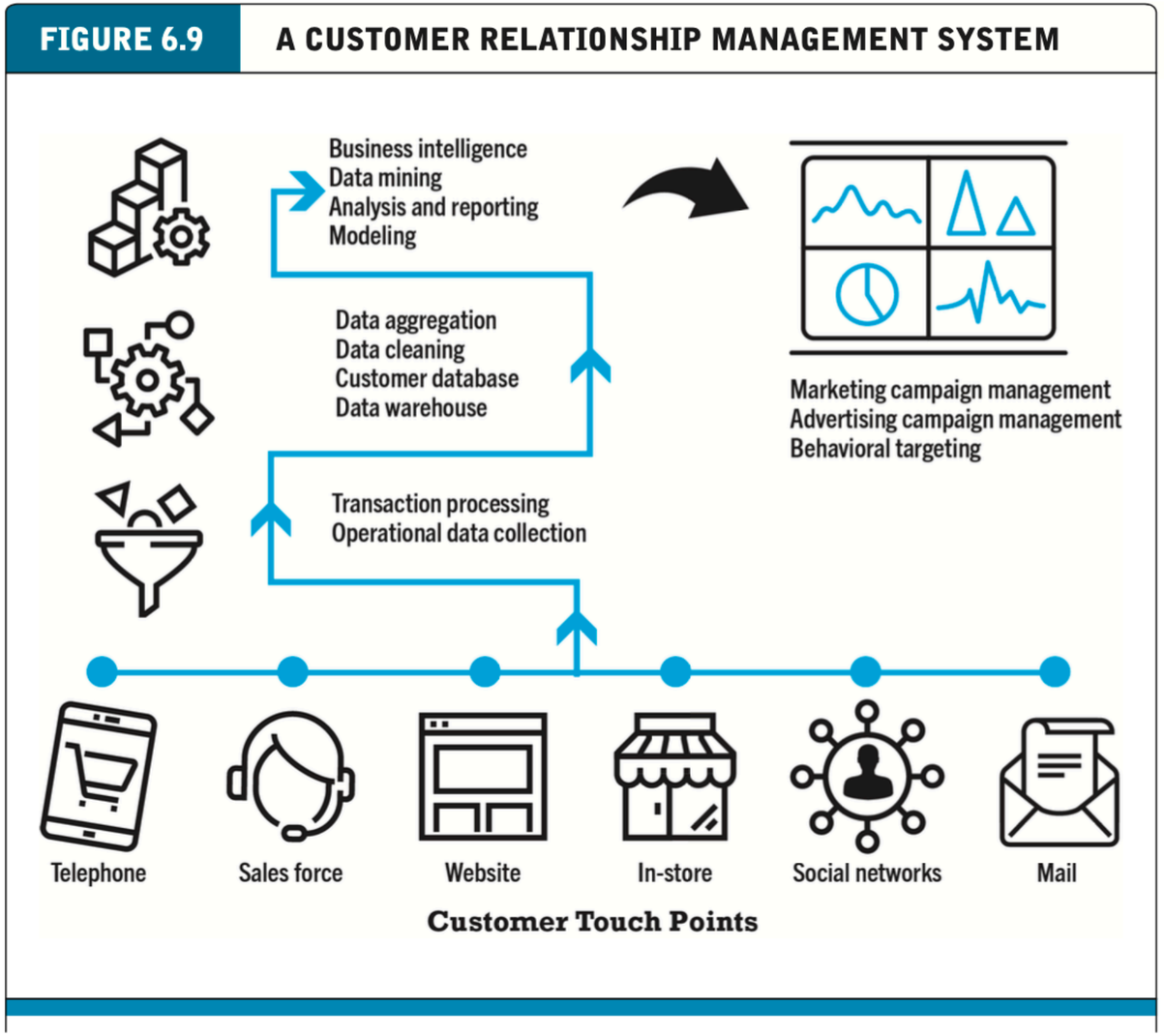
MARKETING AUTOMATION AND CUSTOMER RELATIONSHIP MANAGEMENT (CRM) SYSTEMS

A **customer relationship management** (**CRM**) **system** is a repository of customer information that records all of the contacts that a customer has with a firm (including websites) and generates a customer profile available to everyone in the firm with a need to "know the customer." CRM systems also supply the analytical software required to analyze and use customer information.

Customers come to firms not just over the Web but also through telephone call centers, customer service representatives, sales representatives, automated voice response systems, ATMs and kiosks, in-store point-of-sale terminals, and mobile devices (m-commerce). Collectively, these are referred to as "**customer touchpoints**."

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FIGURE 6.9 A CUSTOMER RELATIONSHIP MANAGEMENT SYSTEM



Understanding the Cost and Benefits of Online Marketing Communication

As we noted earlier, online marketing communications still comprise only a small part of the total marketing communications universe. While there are several reasons why this is the case, two of the main ones are concerns about how well online advertising really works and about how to adequately measure the costs and benefits of online advertising.



Online Marketing Metrics : Lexicon

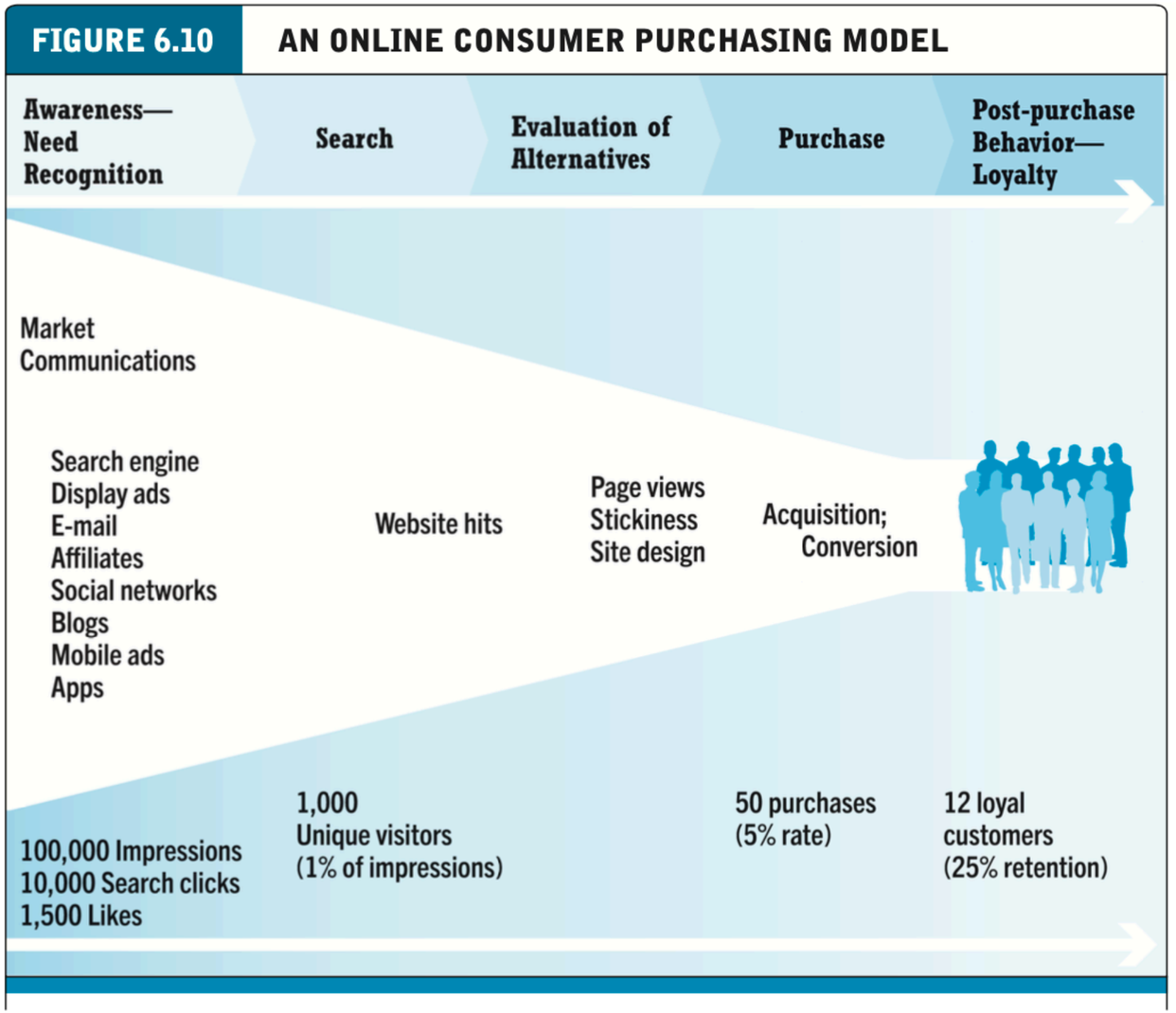
In order to understand the process of attracting prospects via marketing communications and converting them into customers, you will need to be familiar with online marketing terminology.

TABLE 6.6 MARKETING METRICS LEXICON

DISPLAY AD METRICS	DESCRIPTION
Impressions	Number of times an ad is served
Click-through rate (CTR)	Percentage of times an ad is clicked
View-through rate (VTR)	Percentage of times an ad is not clicked immediately but the website is visited within 30 days
Hits	Number of HTTP requests
Page views	Number of pages viewed
Viewability rate	Percentage of ads that are actually seen online
Unique visitors	Number of unique visitors in a period
Loyalty	Measured variously as the number of page views, frequency of single-user visits to the website, or percentage of customers who return to the site in a year to make additional purchases
Reach	Percentage of website visitors who are potential buyers; or the percentage of total market buyers who buy at a site
Recency	Time elapsed since the last action taken by a buyer, such as a website visit or purchase
Stickiness (duration)	Average length of stay at a website
Acquisition rate	Percentage of visitors who indicate an interest in the website's products by registering or visiting product pages
Conversion rate	Percentage of visitors who become customers
Browse-to-buy ratio	Ratio of items purchased to product views
View-to-cart ratio	Ratio of "Add to cart" clicks to product views
Cart conversion rate	Ratio of actual orders to "Add to cart" clicks
Checkout conversion rate	Ratio of actual orders to checkouts started
Abandonment rate	Percentage of shoppers who begin a shopping cart purchase but then leave the website without completing a purchase (similar to checkout conversion rate)
Retention rate	Percentage of existing customers who continue to buy on a regular basis (similar to loyalty)
Attrition rate	Percentage of customers who do not return during the next year after an initial purchase
VIDEO ADVERTISING METRICS	
View time	How long does the ad actually stay in view while it plays
Completion rate	How many viewers watched the complete video
Skip rate	How many viewers skipped the video
E-MAIL METRICS	
Open rate	Percentage of e-mail recipients who open the e-mail and are exposed to the message
Delivery rate	Percentage of e-mail recipients who received the e-mail
Click-through rate (e-mail)	Percentage of recipients who clicked through to offers
Bounce-back rate	Percentage of e-mails that could not be delivered
Unsubscribe rate	Percentage of recipients who click unsubscribe
Conversion rate (e-mail)	Percentage of recipients who actually buy

Online Marketing Metrics : Lexicon

FIGURE 6.10 AN ONLINE CONSUMER PURCHASING MODEL



How Well Does Online Advertising Work?

What is the most effective kind of online advertising? How does online advertising com- pare to offline advertising? The answers depend on the goals of the campaign, the nature of the product, and the quality of the website you direct customers toward.

TABLE 6.7 ONLINE MARKETING COMMUNICATIONS: TYPICAL CLICK-THROUGH RATES

MARKETING METHODS	TYPICAL CLICKTHROUGH RATES
Banner ads	.05%
Google enhanced search ads (Product Listing Ads)	2.8%–3.6%
Google Ads/search listing	1.55%–2.88%
Google Ads/display ads	.47%–.80%
Video	.15%–.45%
Rich media	.02%–.35%
Sponsorships	1.5%–3.0%
Native ads	.80%–1.8%
Content marketing	.40%–2.35%
Affiliate relationships	.20%–.40%
E-mail marketing in-house list	3.0%–5.0%
E-mail marketing purchased list	.01%–1.5%
Facebook News Feed ads	1.11%–1.5%
Facebook Right Hand ads	.08%–.16%
Instagram Feed Ads	.22%–.88%
Twitter ads	.86%–2.2%
Mobile display ads	.09%–1.25%

The Costs of Online Advertising

According to the Interactive Advertising Bureau, in 2019, about 63% of digital ad spending used a performance-based pricing model, while 35% used a CPM-based model, and just 1.9% used a hybrid model.

TABLE 6.8 DIFFERENT PRICING MODELS FOR ONLINE ADVERTISEMENTS

PRICING MODEL	DESCRIPTION
Barter	Exchange of ad space for something of equal value
Cost per thousand (CPM)	Advertiser pays for impressions in 1,000-unit lots
Cost per click (CPC)	Advertiser pays prenegotiated fee for each click ad received
Cost per lead (CPL)	Advertiser pays only for qualified leads or contacts
Cost per action (CPA)	Advertiser pays only for those users who perform a specific action, such as registering, purchasing, etc.
Hybrid	Two or more of these models used together
Sponsorship	Term-based; advertiser pays fixed fee for a slot on a website

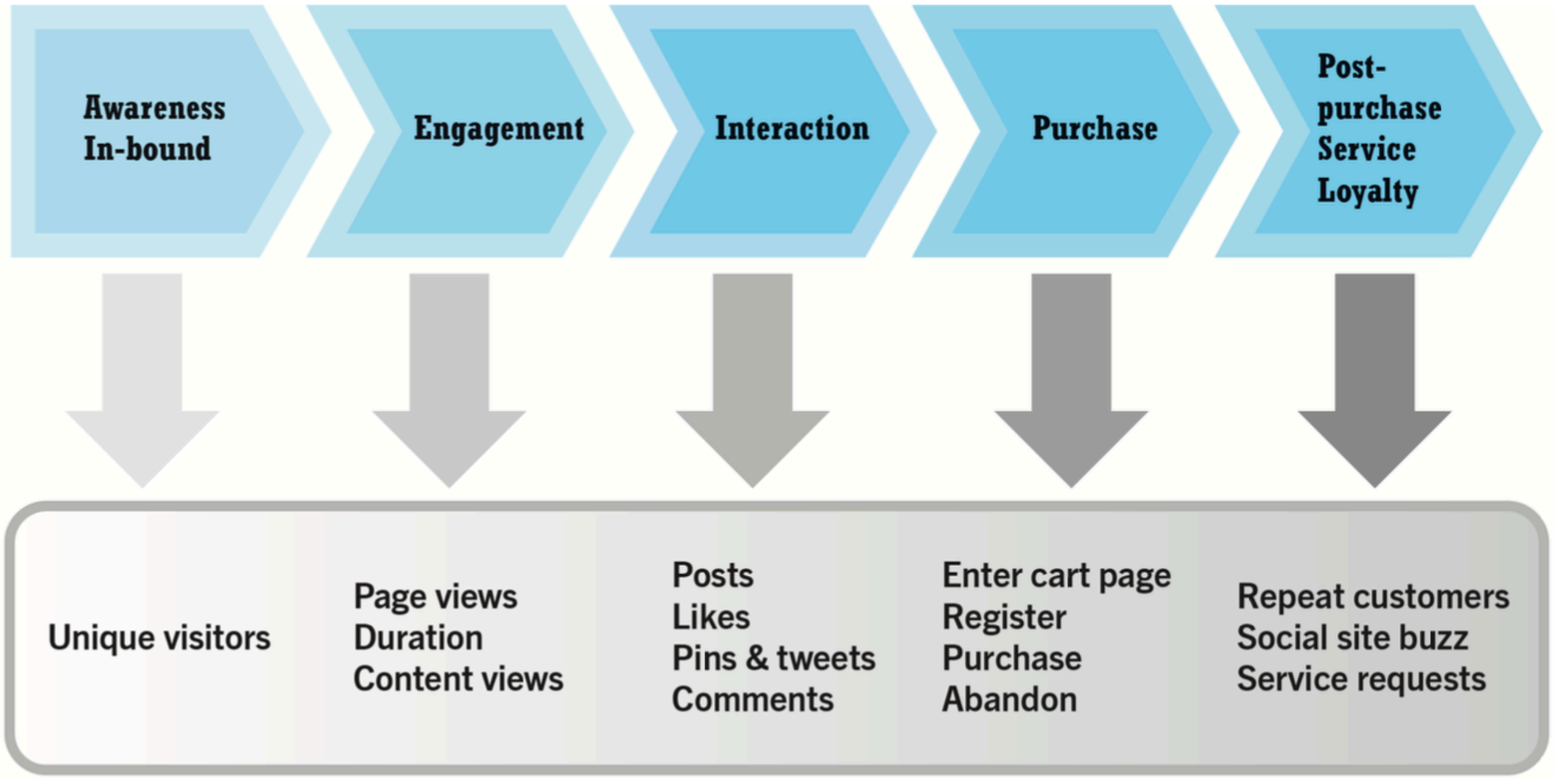
Marketing Analytics : Software for Measuring Online Marketing Results

A number of software programs are available to automatically calculate activities at a website or on a mobile device. Tracking the viewing and behavior of consumers across myriad devices and media channels is a much more difficult task. Other software programs and services assist marketing managers in identifying exactly which marketing initiatives are paying off and which are not.

The purpose of marketing is to convert shoppers into customers who purchase what you sell. The process of converting shoppers into customers is often called a “purchasing funnel.” We have characterized this as a process, rather than a funnel, that is composed of several stages: awareness, engagement, interaction, purchase, and post-purchase service and loyalty. Marketing analytics software collects, stores, analyzes, and graphically presents data on each of the stages in the conversion of shoppers to customers.

Marketing Analytics : Software for Measuring Online Marketing Results

FIGURE 6.11 MARKETING ANALYTICS AND THE ONLINE PURCHASING PROCESS



Do you have any questions?

มีคำถามหรือข้อสงสัยเกี่ยวกับเนื้อหาที่เรียนในวันนี้





Next Session Topic

หัวข้อสำหรับสัปดาห์หน้า

- Social, Mobile and Local Marketing for E-Commerce